
Algorithm 1: CTM Voting

Input: OCR frames $\mathcal{P}$, templates $\mathcal{T}$, thresholds τ_s, τ_c

Output: fused plate $\hat{y}$

$\mathcal{C} \leftarrow \emptyset$;

foreach $P_i \in \mathcal{P}$ **do**

$x_i \leftarrow \text{ExtractTokens}(\text{Normalize}(P_i))$;

if $x_i \neq \emptyset$ **then**

$(T_i, s_i) \leftarrow \arg \max_{T \in \mathcal{T}} \text{Score}(x_i, T)$;

$\mathcal{C} \leftarrow \mathcal{C} \cup \{(x_i, T_i, s_i)\}$;

if $\mathcal{C} = \emptyset$ **then**

return $\emptyset$;

$T^* \leftarrow \text{DominantTemplate}(\mathcal{C})$;

$A \leftarrow \{\text{Align}(x_i, T^*) \mid (x_i, T_i, s_i) \in \mathcal{C}\}$;

foreach slot j of T^* **do**

$V_j(c) \leftarrow \{p \mid A_{ij} = (c, p)\}$;

$c^* \leftarrow \arg \max_c (|V_j(c)|, \sum V_j(c))$;

$\rho \leftarrow |V_j(c^*)| / \sum_c |V_j(c)|$, $\bar{p} \leftarrow \text{mean}(V_j(c^*))$;

$\hat{c}_j \leftarrow \begin{cases} c^*, & \rho \geq \tau_s \wedge \bar{p} \geq \tau_c; \\ ?, & \text{otherwise} \end{cases}$;

return $\text{Render}(T^*, \{\hat{c}_j\})$;
